

charge cycle will be extended.

Output-voltage regulation accuracy. The output-voltage regulation accuracy is critical in order to obtain the desired goal: maximise the battery capacity usage. A small decrease in output voltage accuracy results in a large decrease in capacity. However, the output voltage cannot be set arbitrarily high because of safety and reliability concerns.

Fig. 2 depicts the importance of output-voltage regulation accuracy.

Charge termination method. It cannot be stressed enough that over-charging is the Achilles' heel of Li-ion cells. Accurate charge termination methods are essential for a safe, reliable, charging system.

Cell temperature monitoring. The temperature range over which a Li-ion battery should be charged is 0°C to 45°C, typically. Charging the battery at temperatures outside of this range may cause the battery to become hot. During a charge cycle, the pressure inside the battery increases causing the battery to swell. Temperature and pressure are directly related. As the temperature rises, the pressure can become excessive. This can lead to a mechanical breakdown inside the battery or venting. Charging the battery outside of this temperature range may also harm the performance of the battery or reduce the battery's life expectancy.

Generally, thermistors are included in Lithium-ion battery packs in order to accurately measure the battery temperature. The charger measures the resistance value of the thermistor between the thermistor terminal and the negative terminal. Charging is inhibited when the resistance, and therefore the temperature, is outside the specified operating range.

Battery discharge current or reverse leakage current. In many applications, the charging system remains connected to the battery in the absence of input power. The charging system should minimise the current drain from the battery when input power is not

present. The maximum current drain should be below a few microamperes and, typically, below one microampere.

Application examples

Taking the above system considerations into account, an appropriate charge management system can be developed.

Linear solutions. Linear charging solutions are generally employed when a well-regulated input source is available. Linear solutions, in these applications, offer advantages of ease of use, size and cost.

Due to the low efficiency of a linear charging solution, the most important factor is the thermal design. The thermal design is a direct function of the input voltage, charge current and thermal impedance between the pass transistor and the ambient cooling air. The worst-case situation is when the device transitions from the trickle charge stage to the constant-current stage. In this situation, the pass transistor has to dissipate the maximum power. A trade-off must be made between the charge current, size, cost and thermal requirements of the charging system.

Take, for example, an application required to charge a 1000mAh, single Li-ion cell from a 5V±5% input at a constant-current charge rate of 0.5C or 1C. Fig. 3 depicts a low-cost, standalone linear solution. With a few external components, the preferred charge algorithm is implemented. The MCP73843 combines high-accuracy constant-current, constant-voltage regulation with automatic charge termination.

In an effort to further reduce the size, cost and complexity of linear solutions, many of the external components can be integrated into the charge management controller. Advanced packaging and reduced flexibility come along with higher integration. These packages require advanced equipment for manufacturing, and, in many instances, preclude rework.

TRONTEK

**RECHARGEABLE
SEALED
MAINTENANCE
FREE LEAD
ACID BATTERIES**



12V 7.5Ah

6v/4.5ah to 12v/100ah.

(2v & 4v Models Available on request)

**Preferred by leading OEMS in
India and All over World**

Trontek SMF Batteries

(Manufactured in the most modern & contemporary Taiwan-Made Plant)

Have all the winning features:

- Lead-Calcium grid for extended life
- High discharge rate capacity
- Deep discharge recovery
- Low self discharge & long service life
- No memory effect & high performance battery
- Long life & hassle free power

Marketed In India By :-

Trontek Electronics Pvt Ltd.

S-261, Rajinder Place, New Delhi-110008

Ph.: 91-11-64500900, 45572222

Mob - 9211605333, 9871280808

Fax- 011- 25481607

E-mail - Sales@trontekindia.com,

Website - www.trontekindia.com

TAIWAN:

TRONTEK ELECTRONIC TECH CO. LTD.

4F. No. 163, Shing Ming Road, Nei-hu, Taipei

Taiwan, R.O.C. 114

Tel: 886 2 27936715, 886 2 27942396

Fax: 886 2 27936716

OEMS / DEALERS ENQUIRY SOLICITED

www.trontekindia.com

TRONTEK GROUP